

A padding proof of a conjectured permanent inequality

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Abstract

The unrestricted-cut permanent inequality stated as Conjecture 9.3 by Pan, Skandera, and Wang follows from their balanced inequality, Theorem 8.18, by adjoining an identity matrix. We give the complete reduction, including singular matrices and endpoint cuts. The only non-elementary input is the balanced theorem from their preprint.

For an integer $n \geq 0$, write $[n] = \{1, \dots, n\}$, let $\mathcal{O}_n$ and $\mathcal{E}_n$ be its odd and even elements, and write $A[S] = A_{S,S}$ for a principal submatrix in increasing index order. The empty permanent is 1. A real matrix is *totally nonnegative* (TNN) if every square minor is nonnegative.

We use the following even-order instance of Pan–Skandera–Wang’s Theorem 8.18 [1]: for every $m \geq 2$ and every TNN matrix B of order $2m$,

$$\text{per } B[[m]] \text{ per } B[\{m+1, \dots, 2m\}] \geq \text{per } B[\mathcal{O}_{2m}] \text{ per } B[\mathcal{E}_{2m}]. \quad (1)$$

For $m = 1$ this is an equality.

Theorem 1. *For every integer $n \geq 1$, every TNN matrix $A \in \mathbb{R}^{n \times n}$, and every integer $0 \leq h \leq n$,*

$$\text{per } A[[h]] \text{ per } A[\{h+1, \dots, n\}] \geq \text{per } A[\mathcal{O}_n] \text{ per } A[\mathcal{E}_n]. \quad (2)$$

In particular, Conjecture 9.3 of [1] holds.

Proof. First recall two elementary facts about ordered direct sums. If C, D are TNN, then $C \oplus D$ is TNN: a square minor selecting unequal numbers of first-block rows and first-block columns is zero by rank deficiency, while a minor selecting equal numbers factors as the product of a minor of C and a minor of D , without a sign change. Also,

$$\text{per}(C \oplus D) = \text{per}(C) \text{per}(D), \quad (3)$$

because only permutations preserving both blocks can contribute nonzero summands. These assertions include empty blocks. Every principal submatrix of an identity matrix has permanent 1.

Suppose $0 < h < n$ and $2h \leq n$. Put $d = n - 2h$, $m = n - h$, and

$$B = I_d \oplus A.$$

This is a TNN matrix of order $2m$, with balanced leading and trailing principal blocks $I_d \oplus A[[h]]$ and $A[\{h+1, \dots, n\}]$. Thus the left side of (1) equals the left side of (2). An index i of A becomes $d+i$ in B . Consequently the two parity principal submatrices of B are identity blocks

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direct-summed with $A[\mathcal{O}_n]$ and $A[\mathcal{E}_n]$, in the same order if d is even and the opposite order if d is odd. By (3),

$$\text{per } B[\mathcal{O}_{2m}] \text{ per } B[\mathcal{E}_{2m}] = \text{per } A[\mathcal{O}_n] \text{ per } A[\mathcal{E}_n].$$

Applying (1) proves the claim. The possible case $m = 1$ is $n = 2, h = 1$, where the claim is an equality.

If instead $0 < h < n$ and $2h \geq n$, put $d = 2h - n$, $m = h$, and $B = A \oplus I_d$. The two balanced blocks are now $A[[h]]$ and $A[\{h + 1, \dots, n\}] \oplus I_d$. The original indices of A do not shift, and the parity principal submatrices simply acquire identity direct-summands. Thus both product equalities again hold, and (1) applies. This also includes $d = 0$.

Finally, if $h = 0$ or $h = n$, the left side is $\text{per } A$. Every entry of A is nonnegative. The permutations preserving $\mathcal{O}_n$ and $\mathcal{E}_n$ contribute exactly the product on the right side, which is therefore at most the sum over all permutations. This handles the endpoint cuts and $n = 1$. $\square$

Scope and attribution. The proof uses no computation and no strict-positivity, irreducibility, or nonvanishing assumption. It is a short consequence of the balanced theorem in [1], not an independent proof of that theorem. The cited version explicitly states the balanced result as Theorem 8.18 and the unrestricted-cut assertion as Conjecture 9.3. Since the balanced case is included in the unrestricted assertion, identity padding shows that the two universal statements are equivalent.

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References

- [1] S. Pan, M. Skandera, and J. Wang, *Permanental inequalities and unit interval orders*, arXiv:2610.04809v1 [math.CO], 3 October 2026, Theorem 8.18 and Conjecture 9.3. arXiv:2610.04809v1.